

Modeling Climate Anomalies with Statistical Analysis

Today's goals

Module 1

Introduction to Python for Data Analysis

Course Introduction

-  Course Updates and Accessibility Support
Reading • 1 min
-  Earn Academic Credit for your Work!
Reading • 10 min
-  Course Support
Reading • 10 min
-  Introduction to the Course
Video • 2 min
-  Meet the Instructor
Video • 1 min
-  Introduction to Pandas for Data Loading and
Exploration
Video • 25 min
-  **Pandas**
Reading • 15 min
-  Course Assignments with Jupyter
Reading • 10 min

Visualizing Data with Python

- Introduction to Matplotlib and Seaborn for Data Visualization
Video • 25 min

Module Assignment

- Introduction to Pandas and Matplotlib
Programming Assignment • 30 min
- Pandas and Matplotlib
Graded Assignment • 15 min

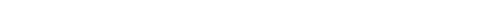
Module 2

Collecting Climate Data

Module 3

Visualizing & Analyzing Climate Data

Pandas

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Pandas is a powerful and versatile open-source data analysis and manipulation library for Python. It provides data structures and functions needed to efficiently manipulate large datasets, making it an essential tool for data scientists, analysts, and anyone working with structured data.

Key Features of Pandas:

1. **Data Structures:**
 - **Series:** A one-dimensional labeled array capable of holding any data type.
 - **DataFrame:** A two-dimensional labeled data structure with columns of potentially different types, similar to a table in a database or an Excel spreadsheet.
2. **Data Manipulation:**
 - **Data Cleaning:** Handling missing data, cleaning up data formats, and transforming data types.
 - **Data Selection:** Extracting specific rows and columns, slicing, filtering, and indexing.
 - **Aggregation and Grouping:** Summarizing data, grouping by specific columns, and applying aggregation functions.
3. **Data Analysis:**
 - **Descriptive Statistics:** Calculating mean, median, mode, standard deviation, and other statistical measures.
 - **Data Transformation:** Applying functions to columns, creating new columns, and reshaping data.
4. **Data Input and Output:**
 - **Reading and Writing Files:** Importing and exporting data to and from various file formats, such as CSV, Excel, SQL databases, and more.
5. **Time Series Analysis:**
 - Handling and manipulating time series data, including date-time indexing and resampling.

Pandas is built on top of NumPy, which provides support for large multi-dimensional arrays and matrices. This foundation ensures that Pandas operations are efficient and optimized for performance.

Getting Started with Pandas

To start using Pandas, you need to install it and import Pandas into your Python script. This will be done for you on Coursera in your assignments. Here's a simple example of creating a DataFrame and performing basic operations:

```
1 import pandas as pd
2
3 # Creating a DataFrame
4 data = {
5     'Name': ['Alice', 'Bob', 'Charlie', 'David'],
6     'Age': [25, 30, 35, 40],
7     'City': ['New York', 'Los Angeles', 'Chicago', 'Houston']
8 }
9 df = pd.DataFrame(data)
10
11 # Displaying the DataFrame
12 print(df)
13
14 # Selecting a column
15 print(df['Age'])
16
17 # Filtering rows
18 print(df[df['Age'] > 30])
19
20 # Adding a new column
21 df['Salary'] = [70000, 80000, 120000, 110000]
22 print(df)
```

To learn more about Pandas see the following user guides, all open in a new tab and are external links:

1. [10 minutes to pandas](#) 
2. [Reshaping and pivot tables](#) 
3. [Merge, join, concatenate and compare](#) 
4. [Group by: split-apply-combine](#) 

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